

ENERGY AI HACKATHON 2026

Participant Problem Statement

Read this document in full before writing any code. It defines the problem, the dataset, the evaluation rules, and what you must present.

1 Problem Formulation

You are given a residential site in Italy connected to the utility grid, a rooftop solar PV system, and a lithium battery. The site electrical load must be fully served at every 15-minute interval without exception.

At each timestep, energy can come from the grid (purchased at a time-varying buy price), from the PV system when generating, or from the battery when discharging. If there is a surplus after the load is served, the excess may charge the battery, be exported to the grid at the prevailing sell price, or both.

Your challenge is to build two things: a load forecasting model, and a rolling-horizon battery dispatch controller that reduces the site electricity bill compared to the defined baselines.

Power Sign Convention

Use these conventions throughout your code, equations, and presentation:

Variable	Sign	Meaning
P_battery	Negative (< 0)	Battery charging
P_battery	Positive (> 0)	Battery discharging
P_grid	Positive (> 0)	Importing from grid
P_grid	Negative (< 0)	Exporting to grid

2 System Specifications

Parameter	Value	Notes
Solar PV capacity	9 kWp	Peak power at standard test conditions
Battery usable capacity	16 kWh	Total usable energy
Battery max charge / discharge power	8 kW	Symmetric limit for charge and discharge
Battery round-trip efficiency	90 %	Apply $\sqrt{0.90} \approx 0.9487$ per direction
Grid connection limit	6 kW	Maximum instantaneous import and export
Initial battery SoC	50 %	At the start of the 2025 optimization window

Italian Residential Time-of-Use Tariff

Grid electricity follows a three-band residential time-of-use tariff. Buy prices are given below. The sell price at each timestep is provided directly in the dataset.

Band	Buy price (€/kWh)	Active hours
F1	0.2540	Mon–Fri 08:00–19:00 (excluding national holidays)
F2	0.2682	Mon–Fri 07:00–08:00 and 19:00–23:00 Saturday 07:00–23:00 (excl. holidays)
F3	0.2440	Mon–Sat 00:00–07:00 and 23:00–24:00 Sundays and national holidays: all day

Units

Power: kW · Energy: kWh · Prices: €/kWh · Cost and savings: € · Time resolution: 15 min ($\Delta t = 0.25$ h)

3 Dataset

You receive one Excel file with two sheets — 2024 and 2025 — both at 15-minute resolution. The 2025 sheet includes the actual battery power recorded on site by the existing controller.

Column	Unit	Description
timestamp	datetime	15-minute interval start, local Italian time (CET/CEST)
load_kw	kW	Site electrical load (always ≥ 0)
pv_kw	kW	PV generation (always ≥ 0 ; zero at night)
buy_price	€/kWh	Grid import price per timestep (F1 / F2 / F3 band)
sell_price	€/kWh	Grid export price per timestep (from market data)
p_battery_kw	kW	Actual battery power recorded on site. Negative = charging, Positive = discharging. This is the Baseline A signal.

Train / Test Split

- Training: 2024 data. Use this to train and tune your forecasting model.
- Test and optimization: 2025 data. Report all metrics and compute all electricity bills on 2025 only. Do not train or tune on 2025.

Data integrity: The 2025 p_battery_kw column was recorded directly from on-site sensors. A multi-week window contains values that are inconsistent with the site energy balance. Before building your optimizer, reconstruct the battery SoC trajectory from scratch and verify it is physically consistent. Unexplained SoC drift is a sign of corrupted data.

Day 2 — 13:00: A second load dataset from a different residential site will be released. Run your trained forecasting model on it without retraining and report the results.

4 Challenge Structure

The challenge has one required core task and one optional extension.

CORE TASK — Required

1. Train a load forecasting model on the 2024 data.
2. Apply it to generate a 2025 load forecast.
3. Build a rolling-horizon battery dispatch controller using your forecast (see Section 7).
4. Compare your controller's 2025 electricity bill against both baselines (Section 5).
5. Report RMSE, MAE, and NRMSE on the 2025 test set.
6. Oracle gap: re-run your controller with the actual 2025 load values as input (perfect oracle). Report oracle savings, forecast-based savings, and the gap in € and %.

EXTENSION 1 — What is the optimal forecast horizon?

Run the full 2025 optimization for at least three different horizon lengths H (e.g., 4, 24, and 96 timesteps — 1 hour, 6 hours, and 24 hours). Report total savings and computation time for each.

What is the minimum horizon needed to capture most of the savings benefit? Is there a point of diminishing returns? Justify your chosen H .

Report: savings vs. horizon table with your recommended H and rationale.

5 Baselines

Report savings against both baselines. Both must appear in your final results table.

Baseline A — Historical Operation

Compute the 2025 electricity bill using the `p_battery_kw` column from the dataset. This is exactly what the existing on-site controller did. Your controller must beat this — if it does not, the current system outperforms your solution.

Baseline B — Zero-Intelligence

PV powers the load first at every timestep.
Surplus ($PV > load$): exported to the grid at the sell price. Battery not used.
Deficit ($load > PV$): bought from the grid at the buy price. Battery not used.
Battery stays at its initial SoC throughout. No charging or discharging ever occurs.

Baseline B shows the value of having a controllable battery at all. Baseline A shows the value of a smarter controller versus the existing system.

6 Load Forecasting

Train a model on the 2024 load data and produce a 2025 forecast at 15-minute resolution. You may use any algorithm or framework. Forecasting accuracy has a direct financial impact: a better forecast enables a more effective controller.

Evaluation Metrics

All metrics are computed per 15-minute timestep in kW across the full 2025 period.

Metric	Unit	Definition
RMSE	kW	Root mean squared error per timestep. Penalises large errors more than MAE.
MAE	kW	Mean absolute error per timestep.
NRMSE	%	$\text{RMSE} \div \text{mean}(\text{load}) \times 100$. Primary ranking metric — normalised so results are comparable across sites.

Report all three metrics for the 2025 test set and for the Day 2 surprise dataset. NRMSE is the primary forecasting ranking metric.

Practical Notes

- Split 2024 into a training portion and a validation portion before touching 2025 data
- Residential load has strong daily and weekly seasonality — hour-of-day, day-of-week, month, and public holiday flags are effective features
- Temperature data, if sourced externally, can improve accuracy for thermally-sensitive loads
- Validate that your train/test split does not leak future information into the model

7 Battery Dispatch Controller

Rolling-Horizon Requirement

Your controller must operate causally: at each decision step it may only use information available up to that moment. It looks ahead over a finite forecast horizon H , solves for the optimal dispatch over that window, executes only the first decision, then advances one step and repeats.

A one-shot batch optimization over the full 2025 year — where the optimizer simultaneously accesses all future load, PV, and price data — is not acceptable. It does not reflect how a real controller operates and produces artificially inflated savings figures.

This type of controller is commonly referred to as Model Predictive Control (MPC). Reinforcement Learning (RL) is an alternative framework, though significantly more complex to implement well within the hackathon timeframe.

Objective Function

At each horizon window, minimise the total electricity bill:

$$\text{Bill} = \sum [P_{\text{grid}}(t) \times \text{buy_price}(t) \times \mathbb{I}(P_{\text{grid}} > 0) - |P_{\text{grid}}(t)| \times \text{sell_price}(t) \times \mathbb{I}(P_{\text{grid}} < 0)] \times \Delta t$$

The first term is the cost of grid imports; the second is revenue from exports. Each applies only when P_{grid} has the corresponding sign.

Hard Constraints

- Grid power: $-6 \text{ kW} \leq P_{\text{grid}}(t) \leq 6 \text{ kW}$
- Battery power: $-8 \text{ kW} \leq P_{\text{battery}}(t) \leq 8 \text{ kW}$
- State of charge: $0 \leq \text{SoC}(t) \leq 1$

Choice of Optimization Method

You are free to choose how you solve the optimization within each horizon window. Common approaches include rule-based heuristics, linear programming (PuLP, CVXPY, SciPy), and mixed-integer programming. You will be judged on results and on the clarity of your reasoning for your method choice.

Required Results

- 2025 electricity bill under Baseline A (historical) and Baseline B (zero-intelligence)
- 2025 electricity bill under your rolling-horizon controller
- Absolute savings (€) and percentage savings vs. each baseline
- Oracle gap: re-run the optimizer with actual 2025 load as input (perfect oracle). Report oracle savings, forecast-based savings, and the gap in € and %
- Dispatch plot for Week 3 of March 2025: load, PV, P_{battery} , P_{grid} , and SoC

The March Week 3 dispatch plot is mandatory. Judges use it to verify that your controller is behaving physically correctly and that your reported savings are consistent with the dispatch decisions shown.

8 Final Presentation

Each team presents for exactly 3 minutes. A countdown timer is visible throughout. With 30 teams, time is enforced strictly.

Required Slide Structure (max 6 slides)

#	Topic	What judges need to see
1	Forecasting model	Algorithm and key design choices. RMSE, MAE, and NRMSE on the 2025 test set.
2	Controller approach	Optimization method, chosen horizon H, and justification for both.
3	March Week 3 dispatch plot	load, PV, P_battery, P_grid, and SoC for the required week.
4	Results table	Bill: Baseline A, Baseline B, your controller. Absolute (€) and % savings. Oracle gap in € and %. Extension if completed.
5	Generalization	NRMSE on the Day 2 surprise dataset vs. your 2025 result.
6	Hardest problem & next steps	One specific technical problem you faced and how you handled it. What would you improve with one more day?

Submit slides (PDF or PPTX) and code (ZIP or GitHub link) by 15:00 on Day 2.

9 Scoring & Prizes

Scores are awarded by the judging panel from the 3-minute presentation. The extension earns bonus points above the 100-point base.

Base Score (100 points)

Criterion	Pts	How scored
Controller savings vs. Baseline A	35	% savings vs. historical operation on 2025.
Forecasting accuracy — NRMSE on 2025	25	Must report RMSE, MAE, and NRMSE.
Generalization — NRMSE on surprise dataset	25	NRMSE on Day 2 dataset vs. 2025 baseline.
Reasoning & presentation clarity	15	Quality of design justifications and ability to explain controller and model decisions.
TOTAL	100	

Extension Bonus Points

Extension	Bonus	Condition
Extension 1: Horizon sensitivity analysis with recommended H	+5	≥3 horizons tested; results tabulated

Prizes

Place	Prize	Criteria
1st Place	20,000 EGP	Highest composite score
2nd Place	10,000 EGP	Second highest composite score
3rd Place	5,000 EGP	Third highest composite score

Tiebreaker

Ties resolved in order: (1) controller savings, (2) NRMSE on surprise dataset, (3) NRMSE on 2025 test set. If still tied, judges reach consensus.

Disqualification

- Training or tuning the forecasting model on 2025 data
- Submitting a batch optimizer that accesses full 2025 data simultaneously
- Failing to submit slides and code by 15:15 on Day 2